torch.chain_matmul#

https://pytorch.org/docs/stable/generated/torch.chain_matmul.html#torch.chai

Description:

Returns the matrix product of the NNN 2-D tensors. This product is efficiently computed using the matrix chain order algorithm which selects the order in which incurs the lowest cost in terms of arithmetic operations ([CLRS]). Note that since this is a function to compute the product, NNN needs to be greater than or equal to 2; if equal to 2 then a trivial matrix-matrix product is returned. If NNN is 1, then this is a no-op - the original matrix is returned as is. torch.chain_matmul() is deprecated and will be removed in a future PyTorch release. Use torch.linalg.multi_dot() instead, which accepts a list of two or more tensors rather than multiple arguments. matrices (Tensors...) – a sequence of 2 or more 2-D tensors whose product is to be determined. out (Tensor, optional) – the output tensor. Ignored if out = None.

Function Signature

```
torch.chain_matmul(*matrices, out=None)[source]#
Parameters
Returns
Return type
```

Returns:

Tensor

Code Examples

```
>>> a = torch.randn(3, 4)
>>> b = torch.randn(4, 5)
>>> c = torch.randn(5, 6)
>>> d = torch.randn(6, 7)
>>> # will raise a deprecation warning
>>> torch.chain_matmul(a, b, c, d)
tensor([[ -2.3375,  -3.9790,  -4.1119,  -6.6577,   9.5609,
-11.5095,  -3.2614],
        [ 21.4038,   3.3378,  -8.4982,  -5.2457, -10.2561,
-2.4684,   2.7163],
        [ -0.9647,  -5.8917,  -2.3213,  -5.2284,  12.8615,
-12.2816,  -2.5095]])
```

Notes & Warnings

⚠ **WARNING**

Warning torch.chain_matmul() is deprecated and will be removed in a future PyTorch release. Use torch.linalg.multi_dot() instead, which accepts a list of two or more tensors rather than multiple arguments.

Detailed Documentation

Documentation

Returns the matrix product of the NNN 2-D tensors. This product is efficiently computed using the matrix chain order algorithm which selects the order in which incurs the lowest cost in terms of arithmetic operations ([CLRS]). Note that since this is a function to compute the product, NNN needs to be greater than or equal to 2; if equal to 2 then a trivial matrix-matrix product is returned. If NNN is 1, then this is a no-op - the original matrix is returned as is.

torch.chain_matmul() is deprecated and will be removed in a future PyTorch release. Use torch.linalg.multi_dot() instead, which accepts a list of two or more tensors rather than multiple arguments.

matrices (Tensors...) – a sequence of 2 or more 2-D tensors whose product is to be determined.

out (Tensor, optional) – the output tensor. Ignored if out = None.

if the ithi^{th}ith tensor was of dimensions pi×pi+1p_{i} \times p_{i + 1}pi×pi+1, then the product would be of dimensions p1×pN+1p_{1} \times p_{N + 1}p1×pN+1.